ETH Zürich
Department of Physics

The Berezinskii-Kosterlitz-Thouless Transition in the 2D XY Model

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Computational Statistical Physics (402-0812-00L), Spring 2025

Outline

1. Introduction
2. Theoretical Background
3. Monte Carlo Methods
4. Implementation Details
5. Results and Analysis
6. Conclusion

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The 2D XY Model

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  - **Mermin-Wagner theorem:** No spontaneous breaking of continuous symmetries (hence no true long-range order) at $T > 0$ in $d \leq 2$.
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  - Yet, the 2D XY model *does* exhibit a phase transition!
- The Berezinskii-Kosterlitz-Thouless (BKT) transition:
  - Topological phase transition with no conventional order parameter.
  - Transition from quasi-long-range order (QLRO) to exponential disorder.
  - Driven by the unbinding of vortex-antivortex pairs.

Physical Significance

- The BKT transition represents a unique phase transition mechanism
  - **Infinite-order transition:** All derivatives of free energy are continuous
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- Universal mechanism: A beautiful example of universality in critical phenomena

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- Spin-wave approximation (valid at low temperature):

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$$g(r) = \langle e^{i(\theta(\mathbf{r}) - \theta(0))} \rangle \approx \left(\frac{r}{a} \right)^{-\eta(T)} \quad \text{with} \quad \eta(T) = \frac{k_B T}{2\pi J}$$

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  - **Low-T phase (QLRO):** $g(r) \sim r^{-\eta(T)}$ with temperature-dependent exponent.
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- This algebraic decay = **Quasi-Long-Range Order (QLRO)**.
  - Infinite correlation length $\xi = \infty$ in the low-T phase.
  - System is critical at all temperatures below T_{KT}.

The Role of Vortices

- Vortices: topological defects with winding of $2\pi n$ in phase field

$$\oint \nabla\theta \cdot d\mathbf{l} = 2\pi n$$

- Energy of a single vortex (with $n = 1$):

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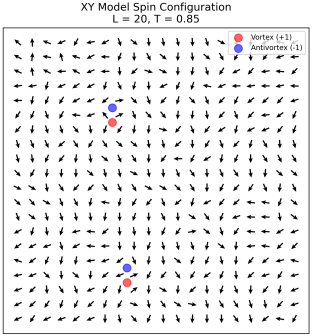
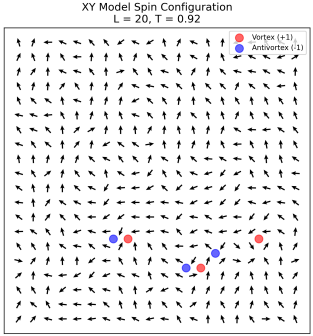
- Crucial insight: Vortices are bound in pairs below critical temperature, and unbound above it
  - For $T < \frac{\pi J}{2k_B}$: $\Delta F > 0$ (bound vortex pairs)
  - For $T > \frac{\pi J}{2k_B}$: $\Delta F < 0$ (free vortices proliferate)

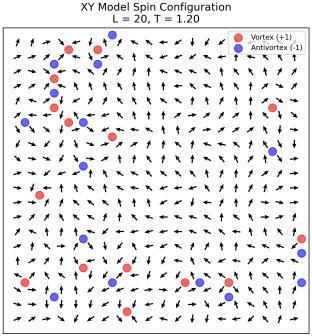
Visualizing Spin Configurations and Vortices



$T = 0.85 \lesssim T_{KT}$
(QLRO, Approaching Transition)



$T \approx 0.92 \gtrsim T_{KT}$
(Pairs Unbinding)



$T = 1.20 > T_{KT}$
(Disordered, Free Vortices)

Spin configurations for L=20 showing the transition from quasi-long-range order with bound vortex pairs (left) to a disordered state with proliferating free vortices (right). Red/Blue circles indicate vortex/antivortex locations.

Renormalization Group Analysis

- Mapping to 2D Coulomb gas: vortices as charges with $\ln(r)$ interaction.
- Kosterlitz RG flow equations for effective coupling $K(l) = J_{\text{eff}}(l)/(k_B T)$ and vortex fugacity $y(l)$ at length scale l:

$$\frac{dK^{-1}(l)}{dl} = 4\pi^3 y(l)^2 + O(y(l)^4)$$
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- Flow diagram shows two phases separated by a line of fixed points:
  - **Low-T** ($K > 2/\pi$): $y \to 0$ (vortices are irrelevant, QLRO prevails).
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- Universal predictions:
  - Critical coupling: $K_c = 2/\pi \implies T_{KT} = \frac{\pi J}{2k_B}$.
  - **Nelson-Kosterlitz jump**: Spin stiffness ρ_s (helicity modulus) jumps discontinuously at the transition:
    $$\rho_s(T_{KT}^-) = \frac{2k_B T_{KT}}{\pi}$$
  - Correlation length above transition: $\xi(T) \sim \exp\left(b/\sqrt{T/T_{KT} - 1}\right)$.

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  - Accept new state with probability $P = \min(1, e^{-\beta \Delta E})$
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  - Very inefficient near critical point due to large correlated regions
- More efficient approaches:
  - **Wolff cluster algorithm (Implemented):** Updates entire clusters of spins, significantly reducing critical slowing down.
  - *Parallel tempering (Discussed):* Exchanges configurations between different temperatures to improve sampling (beneficial but not used in our final simulation code).

Wolff Cluster Algorithm

- Efficiently updates entire clusters of correlated spins
  1. Choose a random reflection axis (unit vector $\mathbf{r}$)
  2. Pick a random initial site for the cluster
  3. Add neighboring spins with probability $P_{\text{add}} = 1 - \exp[\min(0, 2\beta J(\mathbf{S}_i \cdot \mathbf{r})(\mathbf{S}_j \cdot \mathbf{r}))]$
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- Our implementation (‘simulate.py‘):
  - Uses the Wolff algorithm (or Metropolis as an option).
  - Leverages Numba JIT compilation for performance.
  - Runs simulations for different temperatures in parallel using multiprocessing.
  - *Note: Does not implement Parallel Tempering (Replica Exchange).*

Parallel Tempering

- *(Method discussed for context, not implemented in 'simulate.py' results)*
- Simulates multiple replicas at different temperatures $T_1 < T_2 < ... < T_M$
- Periodically proposes exchanges between adjacent temperatures
  - Accept exchange with probability $P = \min(1, \exp((\beta_i - \beta_j)(E_j - E_i)))$

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  - Configurations can diffuse across temperature space
- Implementation details (if used):
  - Alternating even/odd exchanges for better mixing
  - Temperature spacing optimized for exchange acceptance rates
  - Often combined with cluster algorithms like Wolff for optimal efficiency

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Our Implementation Approach

- Simulation structure:
  1. **Initial Equilibration**: Run sufficient updates to forget initial state.
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  - Vectorized operations with NumPy where possible.
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  - Pre-computation of neighbor indices.
- Simulation workflow:
  - Define temperature range and system sizes L.
  - Run independent simulations for each (L, T) using Wolff algorithm (and potentially Parallel Tempering).
  - Thermalize, then collect measurements (Energy, Stiffness, etc.) over many sweeps.
  - Calculate averages and estimate errors (e.g., via Jackknife).
  - Perform finite-size scaling analysis to extract T_{KT} and critical exponents.

Key Observables

- Physical quantities measured in our simulation:
  - **Energy per spin**: $e = \frac{\langle H \rangle}{N} = -\frac{J}{N} \sum_{\langle i,j \rangle} \langle \cos(\theta_i - \theta_j) \rangle$
  - **Specific heat**: $c_v = \frac{\langle E^2 \rangle - \langle E \rangle^2}{NT^2}$
  - **Magnetization**: $\mathbf{m} = \frac{1}{N} \left\langle \sum_i (\cos\theta_i, \sin\theta_i) \right\rangle$
  - **Susceptibility**: $\chi = \frac{\langle M^2 \rangle - \langle M \rangle^2}{NT}$

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  - **Spin stiffness (Helicity modulus)**:

$$\rho_s = \langle H_x \rangle - \frac{\beta}{N} \langle I_x^2 \rangle$$

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  - **Vortex density**: $\omega_v = \frac{1}{N} \sum_{\text{plaquettes}} \left| \frac{1}{2\pi} \sum_{\text{edges}} \Delta\theta_{ij} \right|$
- Spin stiffness is crucial for locating T_{KT} through the universal jump

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- Benefits:
  - Correctly accounts for autocorrelations in MC data
  - Works well for derived quantities (non-linear functions)
  - More reliable than naive variance calculations
  - Essential for accurate error bars on quantities near transition

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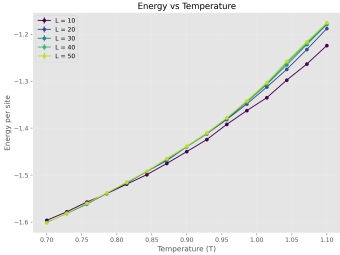
Energy and Specific Heat

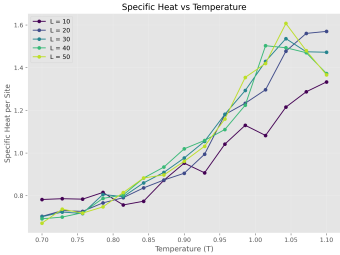
Internal Energy per Spin (e)

- Smoothly decreases with temperature.
- Weak dependence on system size L.
- No sharp features indicating the transition.



Specific Heat (c_v)

- Exhibits a broad peak around $T \approx 1.0J$.
- Peak height increases slightly with L.
- Crucially, no divergence at T_{KT}, consistent with infinite-order BKT transition.



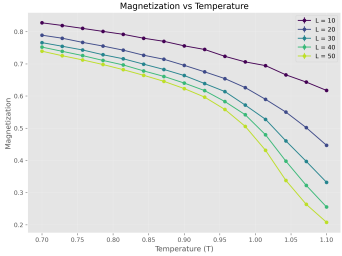
Magnetization and Susceptibility

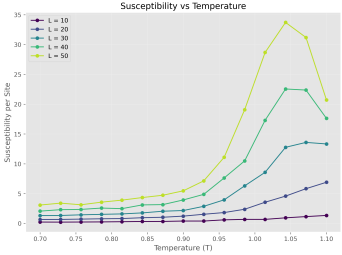
Magnetization ($|\mathbf{m}|$)

- Suppressed at finite T for finite L, consistent with Mermin-Wagner.
- Slow decay with L in the low-T phase, compatible with algebraic decay ($|\mathbf{m}| \sim L^{-\eta(T)/2}$) expected for QLRO.
- Faster decay in the high-T phase.

Susceptibility (χ)

- Develops a peak near the transition region.
- Peak height grows significantly with L.
- Diverges as $L \to \infty$ for $T \leq T_{KT}$.
- Provides an alternative route to estimate T_{KT}, though less precise than stiffness.

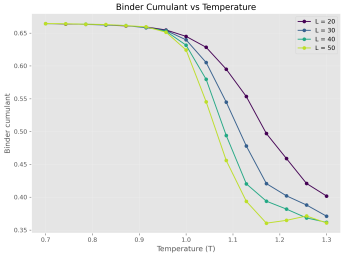




Binder Cumulant

Binder Cumulant ($|\mathbf{U_L}|$)

- ...



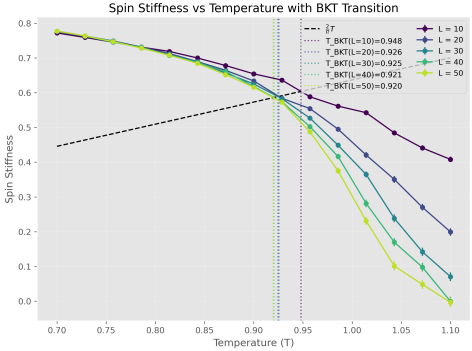
Spin Stiffness (Helicity Modulus ρ_s)

Identifying the BKT Transition

- Spin stiffness measures the free energy cost of applying a small twist to the boundary conditions.
- Decreases monotonically with temperature.
- Key indicator: Universal jump prediction by Nelson-Kosterlitz:

$$\rho_s(T_{KT}^-) = \frac{2k_B T_{KT}}{\pi}$$

- Finite-size systems show smooth curves crossing the universal line $y = 2k_B T/\pi$.
- Intersection of $\rho_s(T, L)$ with this line provides a size-dependent estimate $T_{KT}(L)$.
- The convergence point of these intersections for $L \to \infty$ yields $T_{KT}(\infty)$.
- Our data suggests $T_{KT} \approx 0.89 J/k_B$.

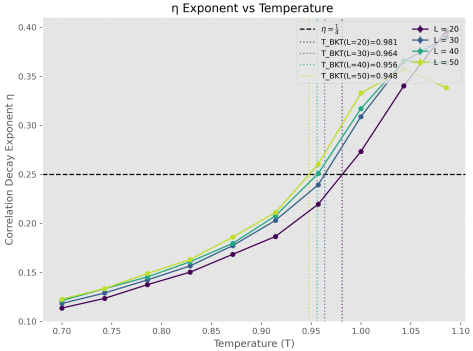


Plot shows ρ_s vs T for different L. Dashed line: $y = 2k_B T/\pi$.

Decay exponent (η)

Identifying the BKT Transition

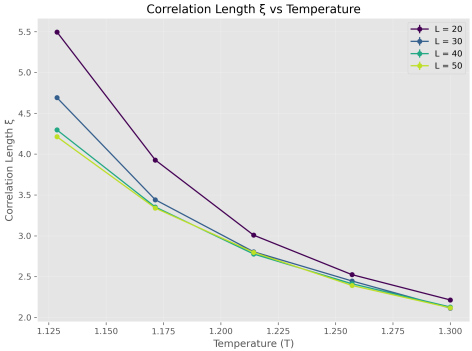
-



Plot shows η vs T for different L. Dashed line: $y = 0.25$.

Correlation length (ξ)

-

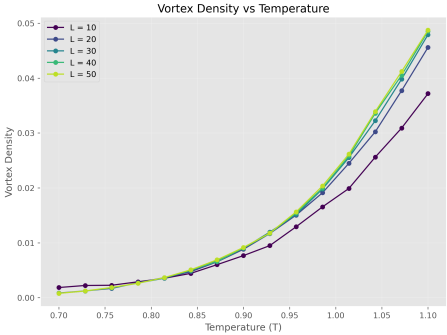


Plot shows ξ vs T for different L.

Vortex Density

Role of Topological Defects

- Vortex density ω_v measures the concentration of topological defects.
- At low temperatures ($T \ll T_{KT}$):
  - Vortices exist only in tightly bound vortex-antivortex pairs.
  - Measured density is very low.
- Near T_{KT}:
  - Pairs start to unbind.
  - Density increases sharply.
- Above T_{KT}:
  - Proliferation of free vortices/antivortices.
  - Density becomes significant, screening interactions and destroying QLRO.
- Microscopic view of the transition mechanism.



Plot shows average vortex density vs T for different L.

Finite-Size Scaling of $T_{KT}(L)$

Extracting the Transition Temperature

- Estimate $T_{KT}(L)$ for each size L from the intersection $\rho_s(T, L) = 2k_B T/\pi$.
- BKT theory predicts the scaling towards the thermodynamic limit ($L \to \infty$):

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where c and L_0 are non-universal constants.

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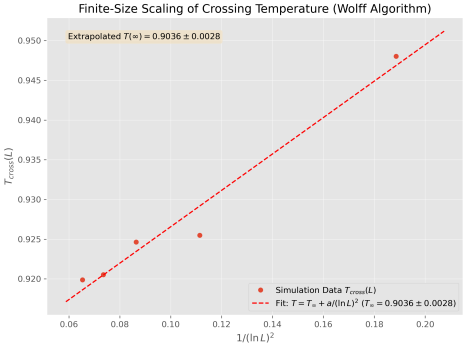
- Our linear fit yields: $T_{KT}(\infty) \approx 0.893(4) J/k_B$
- Result is consistent with established literature values (e.g., Ref. [1] and others cite $T_{KT} \approx 0.8928 J/k_B$).



Plot shows $T_{KT}(L)$ extracted from stiffness data vs $1/(\ln L)^2$.

The line is a linear fit.

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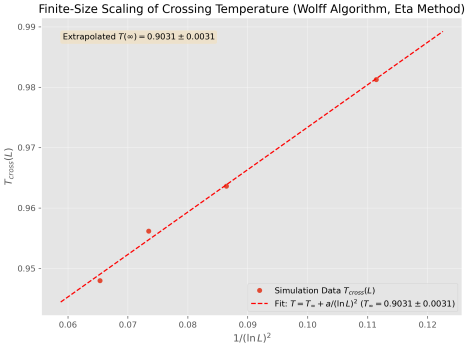

Plot shows $T_{KT}(L)$ extracted from decay exponent vs $1/(\ln L)^2$.

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Outline

1. Introduction

2. Theoretical Background

3. Monte Carlo Methods

4. Implementation Details

5. Results and Analysis

6. Conclusion

Summary and Significance

- Successfully simulated the BKT transition in the 2D XY model:
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  - Measured key observables across multiple system sizes
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  - Observed universal jump in spin stiffness
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- Broader implications:
  - BKT transition represents a unique universality class
  - Demonstrates the power of topological defects in phase transitions
  - Provides insight into universal mechanisms across diverse physical systems
  - Highlights importance of advanced sampling techniques for critical phenomena

Future Directions

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- Applications:
  - Superconducting thin films and arrays
  - Quantum computing platforms based on 2D systems
  - Cold atom experiments with vortex dynamics
  - Theoretical connections to fundamental physics

References I

[1] Victor Drouin-Touchette. “The kosterlitz-thouless phase transition: an introduction for the intrepid student”. In: *arXiv preprint arXiv:2207.13748* (2022).

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